

Buried interface molecular hybrid for inverted perovskite solar cells

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Sanwan Liu^{1,2,13}, Jingbai Li^{3,13}, Wenshan Xiao^{4,13}, Rui Chen^{1,13}, Zhenxing Sun^{1,13}, Yong Zhang^{5,13}, Xia Lei^{3,5}, Shuaifeng Hu⁶, Manuel Kober-Czerny⁶, Jianan Wang¹, Fumeng Ren¹, Qisen Zhou¹, Hasan Raza¹, You Gao¹, Yitong Ji⁴, Sibao Li⁷, Huan Li⁷, Longbin Qiu⁷, Wenchao Huang^{4,8}, Yan Zhao^{9,10}, Baomin Xu⁵, Zonghao Liu^{1,2,13}✉, Henry J. Snaith⁶, Nam-Gyu Park^{11,12}✉ & Wei Chen^{1,2}✉

Perovskite solar cells with an inverted architecture provide a key pathway for commercializing this emerging photovoltaic technology because of the better power conversion efficiency and operational stability compared with the normal device structure. Specifically, power conversion efficiencies of the inverted perovskite solar cells have exceeded 25% owing to the development of improved self-assembled molecules^{1–5} and passivation strategies^{6–8}. However, poor wettability and agglomeration of self-assembled molecules^{9–12} cause interfacial losses, impeding further improvement in the power conversion efficiency and stability. Here we report a molecular hybrid at the buried interface in inverted perovskite solar cells that co-assembled the popular self-assembled molecule [4-(3,6-dimethyl-9*H*-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) with the multiple aromatic carboxylic acid 4,4',4''-nitrotribenzoic acid (NA) to improve the heterojunction interface. The molecular hybrid of Me-4PACz with NA could substantially improve the interfacial characteristics. The resulting inverted perovskite solar cells demonstrated a record certified steady-state efficiency of 26.54%. Crucially, this strategy aligns seamlessly with large-scale manufacturing, achieving one of the highest certified power conversion efficiencies for inverted mini-modules at 22.74% (aperture area 11.1 cm²). Our device also maintained 96.1% of its initial power conversion efficiency after more than 2,400 h of 1-sun operation in ambient air.

Perovskite solar cells (PSCs) with an inverted (p–i–n) architecture are recognized to be one of the mainstream technical routes for the commercialization of this emerging photovoltaic technique owing to their competitive power conversion efficiencies (PCEs), good stability and compatibility with tandem solar cells. Currently, PCEs of these PSCs exceed 25% because of the development of hole-selective layers (HSLs)^{1–4} and passivation strategies^{6–8}. In particular, self-assembled molecules (SAMs)-based HSLs have been widely used in high-efficiency inverted PSCs reported recently because of the advantages of appropriate energy levels for efficient carrier extraction and low non-radiative recombination loss⁴. However, commonly used SAMs, such as [4-(3,6-dimethyl-9*H*-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz), exhibit inhomogeneous distribution^{12,13} on the substrate and poor surface wettability to perovskite precursors^{9–11}. These characteristics challenge the direct deposition of high-quality perovskite film on top of Me-4PACz. These disadvantages will result in a low fabrication yield

and undesired interfacial losses at buried interfaces, especially when manufacturing large-area devices.

Here we report a hybrid SAM strategy, in which the widely used SAM, Me-4PACz, is assembled with a multiple carboxylic acid functionalized aromatic molecule, 4,4',4''-nitrotribenzoic acid (NA), to form mixed SAMs (denoted as NA–Me) on the surface of nickel oxide (NiO) to fabricate inverted PSCs. With NA molecules in the Me-4PACz layer, the wettability of the perovskite solution on the HSL is improved, leading to the reduction of nanovoids and the release of stress at the buried interface. Owing to the interaction between Me-4PACz and the triphenylamine moiety in the NA molecule, NA–Me can reduce the agglomerations of Me-4PACz to homogenize the distribution of Me-4PACz, thus facilitating carrier extraction and reducing non-radiative recombination at the NiO–perovskite interface. The inverted devices based on formamidinium-caesium perovskite with the composition of FA_{0.95}CS_{0.05}PbI₃ achieved optimal PCEs of 26.69% with a high open-circuit

¹Wuhan National Laboratory for Optoelectronics (WNLO), Huazhong University of Science and Technology (HUST), Wuhan, China. ²Optics Valley Laboratory, Wuhan, China. ³Hoffmann Institute of Advanced Materials, Shenzhen Polytechnic University, Shenzhen, China. ⁴Key State Laboratory of Advanced Technology for Materials Synthesis and Processing, School of Materials Science and Engineering, Wuhan University of Technology, Wuhan, China. ⁵Department of Materials Science and Engineering, Southern University of Science and Technology, Shenzhen, China. ⁶Clarendon Laboratory, Department of Physics, University of Oxford, Oxford, UK. ⁷Shenzhen Key Laboratory of Intelligent Robotics and Flexible Manufacturing Systems, Department of Mechanical and Energy Engineering, SUSTech Energy Institute for Carbon Neutrality, Southern University of Science and Technology, Shenzhen, China. ⁸Hubei Longzhong Laboratory, Wuhan University of Technology Xiangyang Demonstration Zone, Xiangyang, China. ⁹College of Materials Science and Engineering, Sichuan University, Chengdu, China. ¹⁰The Institute of Technological Sciences, Wuhan University, Wuhan, China. ¹¹School of Chemical Engineering and Center for Antibonding Regulated Crystals, Sungkyunkwan University (SKKU), Suwon, Republic of Korea. ¹²SKKU Institute of Energy Science and Technology (SIEST), Sungkyunkwan University, Suwon, Republic of Korea. ¹³These authors contributed equally: Sanwan Liu, Jingbai Li, Wenshan Xiao, Rui Chen, Zhenxing Sun, Yong Zhang. ✉e-mail: liuzonghao@hust.edu.cn; npark@skku.edu; wnlochenwei@hust.edu.cn

voltage (V_{oc}) of 1.201 V (certified steady-state efficiency of 26.54%) and 23.06% (certified 22.74%) for small-area PSCs and mini-modules with an aperture area of 11.1 cm², respectively. Furthermore, our devices retained 96.1% and 97.4% of their initial PCEs after operating at 65 °C under 1-sun-equivalent white-light light-emitting diode (LED) illumination in ambient air over 2,400 h and ageing under the damp heat test conditions over 500 h, respectively.

Theoretical screening of hybrid SAMs

To design the hybrid SAMs, we sought to understand the interactions between Me-4PACz (Fig. 1a) and three carboxylic acid functionalized aromatic molecules—that is, NA, benzoic acid (BA) and trimesic acid (TA)—when forming mixed SAM layers between the perovskite and NiO films (Supplementary Fig. 1). We built molecular models for the perovskite/SAMs/NiO heterojunction based on a 10 nm × 10 nm perovskite and NiO surface, filled with 2 nm mixed Me-4PACz and carboxylic acids in 3:1 ratio (Fig. 1b). Computational details can be found in the Methods section and Supplementary Note 1. Molecular dynamics simulations show that Me-4PACz can form dimers, trimers and tetramers in line with the previous report¹², which could cause the formation of nanovoids on the perovskite bottom surface (Fig. 1c). Adding BA did not reduce the agglomeration of Me-4PACz with limited control of the nanovoids (Fig. 1d). By contrast, the adding of NA (Fig. 1e) and TA (Fig. 1f) into Me-4PACz delivered more compact hybrid HSLs. Adjacent to the perovskite surface, the adsorption ratio of carboxylic acid increased from 0.07 to 0.43 following BA < TA < NA (Fig. 1g). It resulted in an increasing coverage ratio on the perovskite surface in the BA–Me, TA–Me and NA–Me hybrids from 0.12 to 0.21 by Me-4PACz and from 0.13 to 0.28 by all SAMs (Fig. 1g and Supplementary Fig. 2). In terms of molecular configuration, density functional theory (DFT) calculations showed the absorbed Me-4PACz lying on the perovskite surface (Fig. 1h) with a preferred configuration for defect passivation. When adsorbed on the NiO surface, both the phosphonic group and π -ring in Me-4PACz could interact with the NiO surface, leading to much higher absorption energy when compared with the three carboxylic acids (Fig. 1i,j and Supplementary Fig. 3). Interestingly, the NA–Me dimer shows notably higher binding energies than other dimers (Fig. 1k) because of the co-existing π – π interactions and hydrogen bonding. Their interaction energies (–0.61 eV) are comparable to the hydrogen binding energy (–0.62 eV) observed in the Me-4PACz tetramer (Supplementary Fig. 4, Supplementary Note 2 and Supplementary Table 1). Thus, the formation of NA–Me dimer could break the Me-4PACz tetramers, which helps reduce its agglomeration to homogenize its distribution, leading to a compact and uniform NiO/SAMs HSL at the buried interface.

Buried interface quality

Furthermore, the introduction of carboxylic acids into the Me-4PACz layer could enhance the wetting of the perovskite precursor on the NiO/SAM substrate (Supplementary Fig. 5), which helped to improve the apparent coverage of the resultant perovskite film (Supplementary Figs. 6 and 7). To further characterize the effect of the wettability of HSLs on the perovskite quality, we used a scatheless peel-off technique to prepare samples for scanning electron microscopy (SEM) characterization of perovskite bottom surface morphology^{3,14} (Fig. 2a–d and Supplementary Fig. 8). Numerous nanovoids were observed at the bottom surface of the perovskite grown on the Me-4PACz SAM because of its hydrophobicity and insufficient wetting. By contrast, the exposed bottom surfaces of the perovskite films deposited on the TA–Me or NA–Me SAMs showed a more compact and smoother film morphology without obvious nanovoids. However, a small number of nanovoids remained in the BA–Me sample. The elimination of nanovoids at the buried interface using the molecular hybrid strategy is further verified by the cross-section SEM characterizations (Fig. 2e–h). Interestingly,

all the perovskite films showed similar morphologies at the top surface (Supplementary Fig. 9).

The improvement in morphologies is correlated with the crystallinity of the perovskite films at the buried region, although the mixed SAMs did not markedly change the bulk crystal quality of the entire perovskite films (Supplementary Fig. 10). To further explore this point, grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements were performed on the exposed bottom surfaces of the perovskite films with an incidence angle of 0.5° (Fig. 2i–l). All GIWAXS scattering results showed that there was no preferential orientation of the perovskite crystals¹. However, the scattering intensity of the (100) peak of the perovskite film deposited on Me-4PACz was significantly lower than those of the three hybrid SAMs. The perovskite film on NA–Me exhibited the strongest intensity in the (100) peak, followed by the perovskite films on the TA–Me and BA–Me SAMs, indicating that the use of mixed SAMs considerably improved the crystal crystallinity of the perovskite films at the buried region.

The notable improvement in the morphology and crystallinity of the perovskite at the bottom region also affects the release of residual stress^{15–17}. We thus investigated the residual stress at the bottom surface of the perovskite films prepared on different HSLs using the grazing incidence X-ray diffraction (GIXRD) technique with the 2θ – $\sin^2\psi$ method with a penetration depth of 50 nm (Supplementary Note 3). It was found that the diffraction peaks of the perovskite film prepared on Me-4PACz, BA–Me and TA–Me SAMs fitted with Gaussian distribution gradually shifted to lower 2θ positions by varying ψ from 5° to 45°, whereas the diffraction peaks of the perovskite film on NA–Me slightly shifted to higher 2θ positions (Fig. 2m,n and Supplementary Fig. 11a,b). The slopes of the fitted lines for the four conditions by fitting the 2θ as a function of $\sin^2\psi$ are shown in Supplementary Fig. 11c. The perovskite films on Me-4PACz, BA–Me and TA–Me substrates exhibited negative slopes, indicating that the films were subjected to tensile stress. By contrast, the slope of the perovskite film on NA–Me showed a diminutive positive value, suggesting a favourable situation with slight compressive stress¹⁸ (Fig. 2o). These results indicate that the hybrid SAM strategy can release the residual tensile stress of perovskite films, which is favourable for the efficiency and stability of PSCs¹⁹.

Reducing interfacial energy loss

Mixed SAMs can be anchored on the NiO surface by their phosphonic acid and carboxylic acid anchoring groups, as demonstrated in the X-ray photoelectron spectrometry (XPS) spectra (Supplementary Fig. 12). The homogeneity of the distribution of three mixed SAMs on the surface of NiO also affects the variation of the surface potential of NiO, as demonstrated in Supplementary Fig. 13. Moreover, the surface potential of the mixed BA–Me and TA–Me samples showed a substantial shift before and after washing, whereas the NA–Me sample did not show a notable shift (Fig. 3a–c). This is because the Me-4PACz has stronger adsorption energy at the surface of NiO when compared with the three carboxylic acids, and the presence of NA facilitates a more homogeneous distribution of Me-4PACz, thereby enhancing the overall binding effect of the mixed NA–Me at the surface of NiO.

We further examined the uniformity of the bottom surface of the perovskite films deposited on different HSLs using laser confocal fluorescence lifetime imaging microscopy. As shown in Fig. 3d–f and Supplementary Fig. 14, the NA–Me-based perovskite film exhibited significantly improved uniformity and enhanced photoluminescence lifetimes over a probed area of 30 μ m × 30 μ m when compared with those of Me-4PACz and TA–Me. This observation indicates that the NA–Me hybrids can effectively improve the uniformity of the perovskite bottom surface and can reduce the non-radiative recombination loss at the buried interface.

The modification of NiO with SAMs could affect the energy-level alignment at the buried interface. The ultraviolet photoelectron

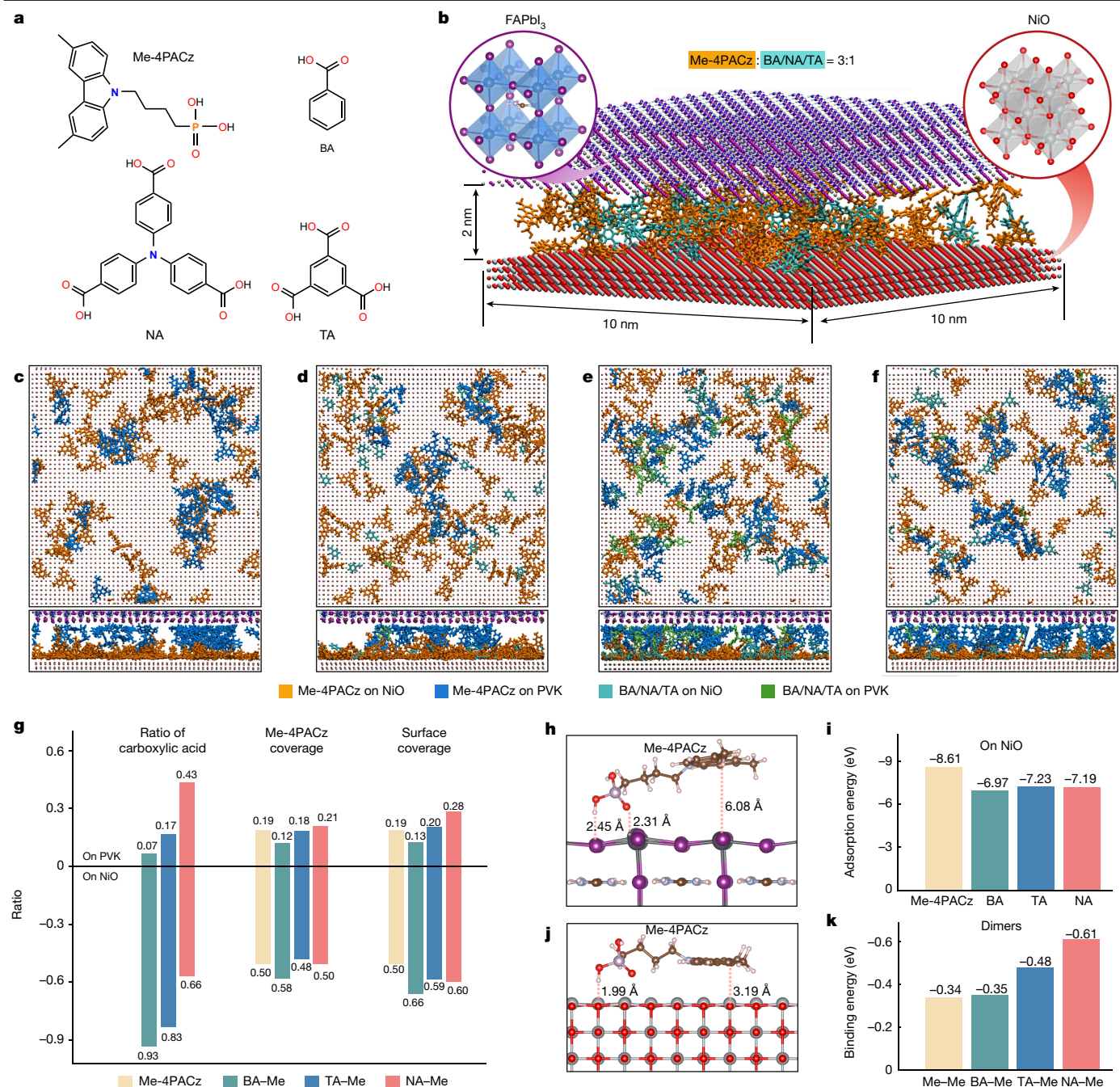


Fig. 1 | Computational simulations of the heterojunction of the HSL.

a, Chemical structure of Me-4PACz and BA, NA and TA. **b**, Illustration of the perovskite/SAMs/NiO heterojunction model for molecular dynamics simulations. **c–f**, Top and side views of equilibrated molecular representations of the heterojunctions with only Me-4PACz (**c**), BA-Me (**d**), NA-Me (**e**) and TA-Me hybrids (**f**). Me-4PACz molecules are shown in orange and blue for the adsorptions on the NiO and perovskite (PVK), respectively. Carboxylic acid functionalized molecules are shown in cyan and green for the adsorptions on the NiO and perovskite (PVK), respectively. Ni and O atoms are shown in grey and red in the background, respectively. The perovskite is omitted in the

top views for clarity. I atoms in the perovskite are highlighted in purple in the side view. **g**, Adsorption analysis of the perovskite and NiO surface in the heterojunction models. The ratio of carboxylic acid functionalized molecules is determined by the number of carboxylic acids within 10 Å of each surface. The Me-4PACz and surface coverage describe the total area covered by Me-4PACz and all SAMs within 5 Å of each surface. **h**, The optimized adsorption structures of Me-4PACz at the NiO surface. **i**, Adsorption energies for SAMs at the NiO surface. **j**, The optimized adsorption structures of Me-4PACz on the perovskite surface. **k**, Binding energies for the typical dimers of SAMs observed in the molecular dynamic simulations. Me-Me is the dimer of Me-4PACz.

spectroscopy (UPS) results (Supplementary Figs. 15 and 16 and Supplementary Table 2) indicate that the energy difference between the NiO and perovskite valance bands for hole extraction is the lowest when modifying NiO with NA-Me (Supplementary Fig. 17). This favourable energy-level alignment was verified to facilitate the hole extraction at the NiO–perovskite interface by the steady-state photoluminescence

and time-resolved photoluminescence (TRPL) measurements (Supplementary Fig. 18, Supplementary Note 4 and Supplementary Table 3). Moreover, the reduced energy loss at the buried interface is also beneficial for improving the V_{oc} of the devices.

To quantify the interfacial non-radiative recombination loss, we measured the photoluminescence quantum yield (PLQY) to estimate

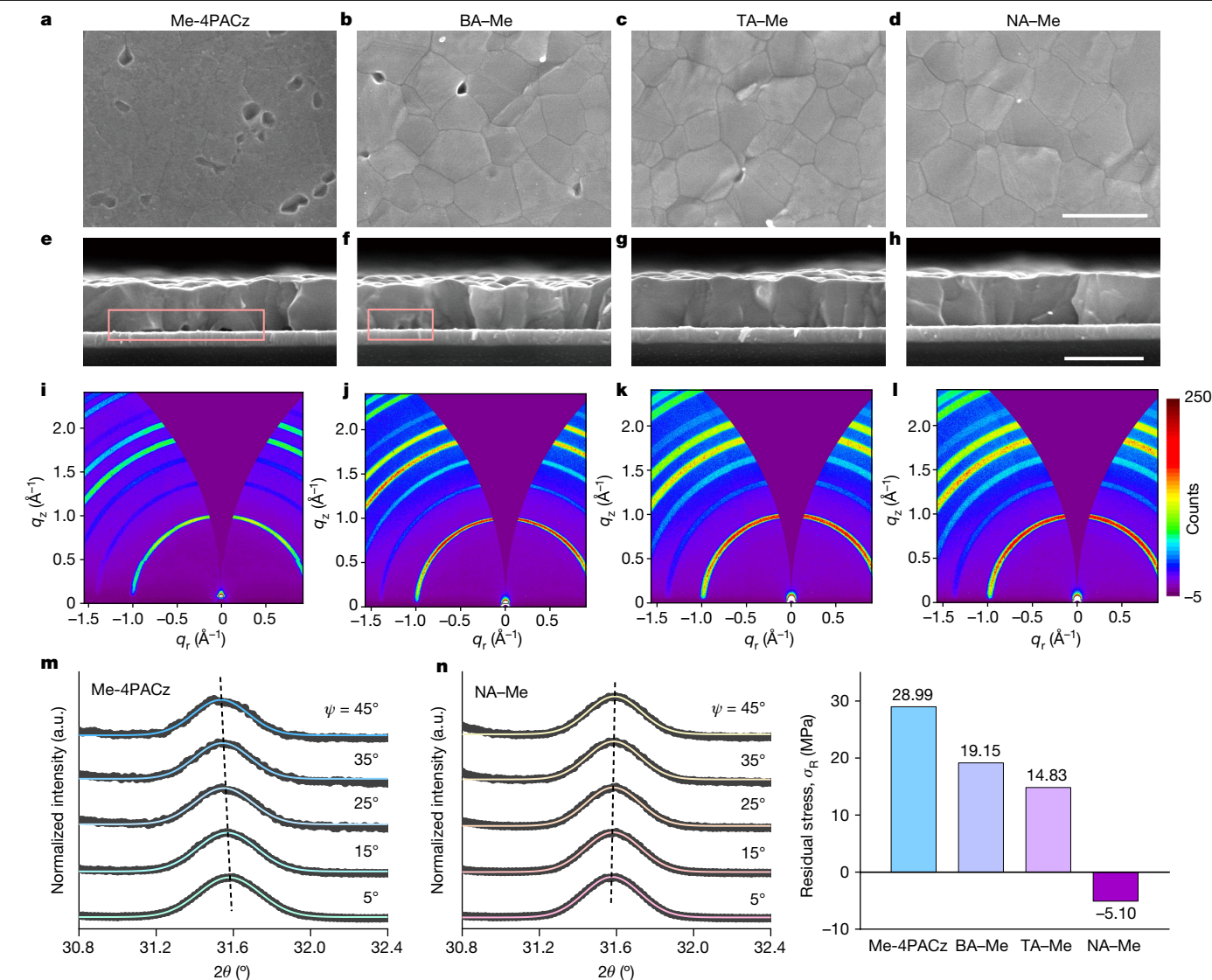


Fig. 2 | The morphology and structure at the perovskite-substrate region. **a–d**, SEM images of the bottom surface of perovskite films on Me-4PACz (**a**), BA-Me (**b**), TA-Me (**c**) and NA-Me (**d**) SAMs. **e–h**, Cross-section SEM images of perovskite films on Me-4PACz (**e**), BA-Me (**f**), TA-Me (**g**) and NA-Me (**h**) SAMs. In **e** and **f**, the pink rectangles point out the nanovoids at the buried interface from the perovskite. **i–l**, Two-dimensional GIWAXS patterns of perovskite films deposited on Me-4PACz (**i**), BA-Me (**j**), TA-Me (**k**) and NA-Me (**l**) SAMs.

q_z and q_r represent the out-of-plane and in-plane scattering vector components, respectively. The colour bar shows the diffraction intensity collected from the GIWAXS detector. **m, n**, GIXRD patterns measured at different ψ angles from 5° to 45° for perovskite films on Me-4PACz (**m**) and NA-Me HSLs (**n**). **o**, Comparison of residual stress in perovskite films on different HSLs. a.u., arbitrary units. Scale bars, 500 nm (**a–d**); 1 μ m (**e–h**).

the quasi-Fermi level splitting in the corresponding samples^{20,21} (Supplementary Note 5). As shown in Fig. 3g and Supplementary Figs. 19–21, the NA-Me sample exhibited the maximum PLQY value of 9.19% and a quasi-Fermi level splitting value of 1.228 eV, leading to the minimal V_{OC} deficit as a result of non-radiation recombination ($\Delta V_{OC}^{non-rad}$) compared with other SAMs. We further used a two-trap-level Shockley–Read–Hall (SRH) model^{22,23} to quantify the effect of NA-Me on the recombination rates of the perovskite films based on the light-intensity-dependent PLQY measurements (Fig. 3h, Supplementary Fig. 22 and Supplementary Table 4). We observe that the PLQY of the NA-Me samples is consistently higher than that of the Me-4PACz samples for the examined photon flux densities. From the fits of the measured PLQY to the SRH model, we determine that the trapping rate into a shallow trap level (k_{tr}) and the non-radiative recombination rate (k_{deep}) are smaller for the NA-Me sample than for the Me-4PACz sample. We interpret this result as the reduced surface recombination at the buried interface for the NA-Me sample and a reduced density of

shallow bulk defects. Details and fitted parameters can be found in Supplementary Note 6.

To further study the passivation mechanism of the hybrid SAMs, we investigated the possible interactions between the hybrid SAMs and the perovskite films by considering four typical perovskite surface defects, I-vacancy (V_I), Pb-vacancy (V_{Pb}) and Pb–I anti-sites (Pb_I and I_{Pb}) using DFT calculations. The bottom perovskite surface was weak p-type corresponding to the approximately stoichiometric Pb and I conditions as indicated by the UPS results^{24,25}. The results of the defect-formation energies calculation are shown in Supplementary Figs. 23–25, Supplementary Note 7 and Supplementary Table 5. The defect transition-level diagram (Fig. 3i) shows that the V_I , V_{Pb} and I_{Pb} defects are thermodynamically more stable than Pb_I near the Fermi level. We thus focused on the V_{Pb}^{2+} defect because it led to notable deep traps (Supplementary Fig. 24). The molecular dynamics simulations of the heterojunction models with a V_{Pb}^{2+} defect show the π -ring of Me-4PACz pointing to the defect in all cases (Fig. 3j, Supplementary

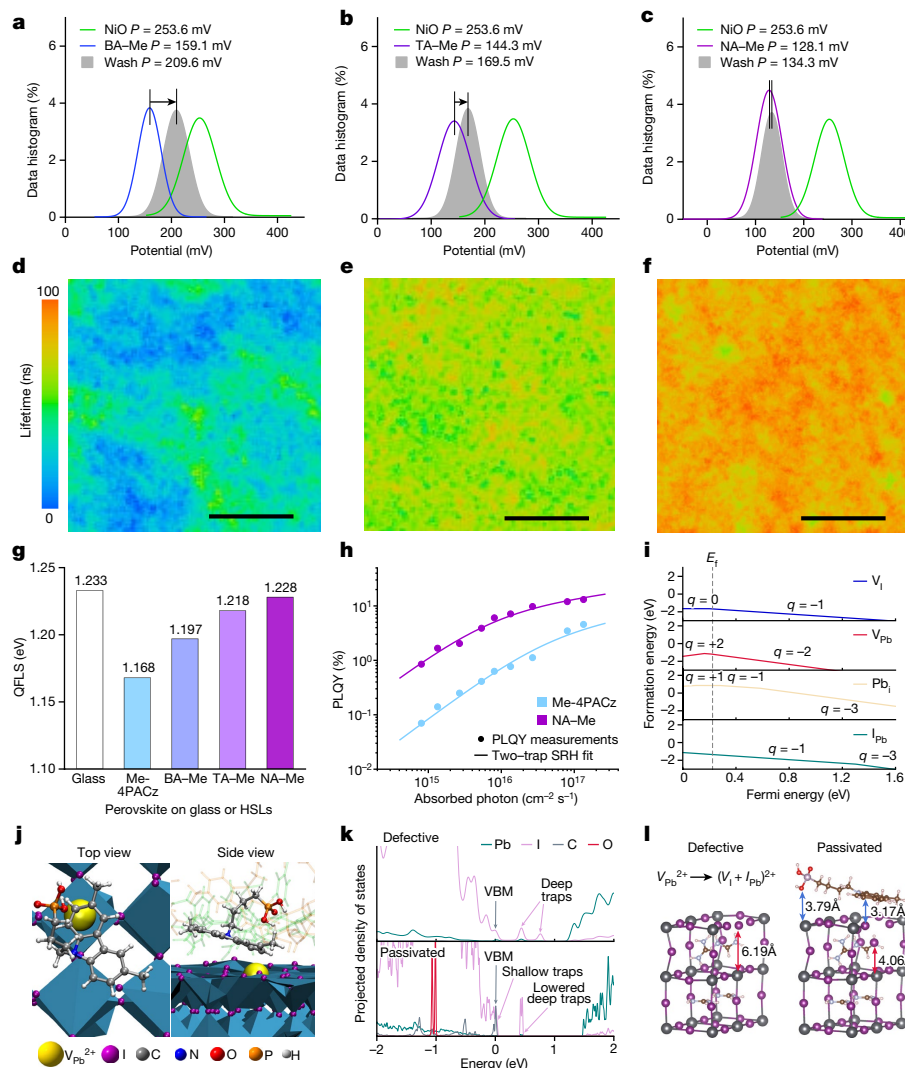


Fig. 3 | Reducing interfacial energy loss. **a–c**, Contact potential difference distributions of different mixed SAMs coated on NiO before and after washing (DMF:DMSO, vol/vol = 4:1): BA-Me (**a**), TA-Me (**b**) and NA-Me (**c**). The black arrow indicates the direction of the change in the surface potential (P) of the sample, specifically that it increases. **d–f**, Fluorescence lifetime imaging microscopy of the exposed bottom surface of perovskite layers on Me-4PACz (**d**), TA-Me (**e**) and NA-Me (**f**). **g**, The calculated quasi-Fermi level splitting (QFLS) of perovskite films on glass and different HSLs. **h**, PLQY measurements dependent on photon

flux densities fitted according to the two-trap-level SRH model. **i**, Defect transition level diagram for perovskite surface defects with charges. **j**, Molecular configuration of Me-4PACz adjacent to the V_{pb}^{2+} defect in the molecular dynamics simulations for the NA-Me hybrid system. **k, l**, Projected density of states (**k**) and $2 \times 2 \times 1$ slice of the optimized structures (**l**) of the V_{pb}^{2+} defect and Me-4PACz passivated perovskite surfaces. Red arrows highlight the broken I–Pb ionic bond; blue arrows indicate the distances between Me-4PACz and the perovskite surface. Scale bars, 10 μm (**d–f**).

Note 8 and Supplementary Figs. 26–30), suggesting an essential role of the π -ring of Me-4PACz in passivating the defect. The projected density of states (Fig. 3k) demonstrated that the π -ring of Me-4PACz passivates the V_{pb}^{2+} deep trap, which could reduce the non-radiative recombination. The V_{pb}^{2+} deep trap resulted from the broken ionic I–Pb bond (6.19 Å), which transformed the Pb-vacancy to an I–Pb anti-site with an I-vacancy (Fig. 3l). The π -ring of Me-4PACz is electron-rich, which could extract the positive charges from the defective surface (Supplementary Fig. 31 and Supplementary Note 9). As a result, the partial restoration of the I–Pb bond (4.06 Å) passivated the V_{pb}^{2+} deep trap.

Device characteristics

We fabricated inverted PSCs with an architecture of indium tin oxide (ITO)/NiO/SAMs/ $\text{FA}_{0.95}\text{Cs}_{0.05}\text{PbI}_3$ /piperazinium iodide (PI) (refs. 26,27)/[6,6]-phenyl-C₆₁-butyric acid methyl ester (PC₆₁BM)/bathocuproine (BCP)/Ag (Supplementary Fig. 32) to evaluate the device performance. The effects of modifying the NiO–perovskite interface with only NA

(TA or BA) on device performance are shown in Supplementary Fig. 33 and Supplementary Note 10. The combination of NiO and Me-4PACz was chosen as the baseline bilayer HSL because it enabled improved efficiencies compared with pristine NiO or Me-4PACz single HSL (Supplementary Fig. 34). The photoelectric parameters of the devices with different HSLs are summarized in Supplementary Table 6. The efficiencies of the devices based on the NA-Me are higher than those with the other two hybrids cases, with the main contribution from the simultaneous enhancements of the V_{OC} and fill factor (FF) (Fig. 4a,b and Supplementary Fig. 35). The current density–voltage (J – V) curves of the champion inverted PSCs with different HSLs are shown in Fig. 4c and Supplementary Fig. 36. The champion device with Me-4PACz gave only a substantial but inferior PCE of 23.29%, with $V_{\text{OC}} = 1.135$ V, $J_{\text{SC}} = 25.80$ mA cm^{−2} and FF = 79.53% (steady-state PCE of 23.02%). Meanwhile, the NA-Me device yielded an optimal efficiency of 26.69% and a high V_{OC} of 1.201 V under reverse scan with low hysteresis (Supplementary Table 7) and exhibited a steady-state PCE of 26.61% (Fig. 4d). Specifically, the J_{SC} of the device is 26.3 mA cm^{−2} under reverse scan,

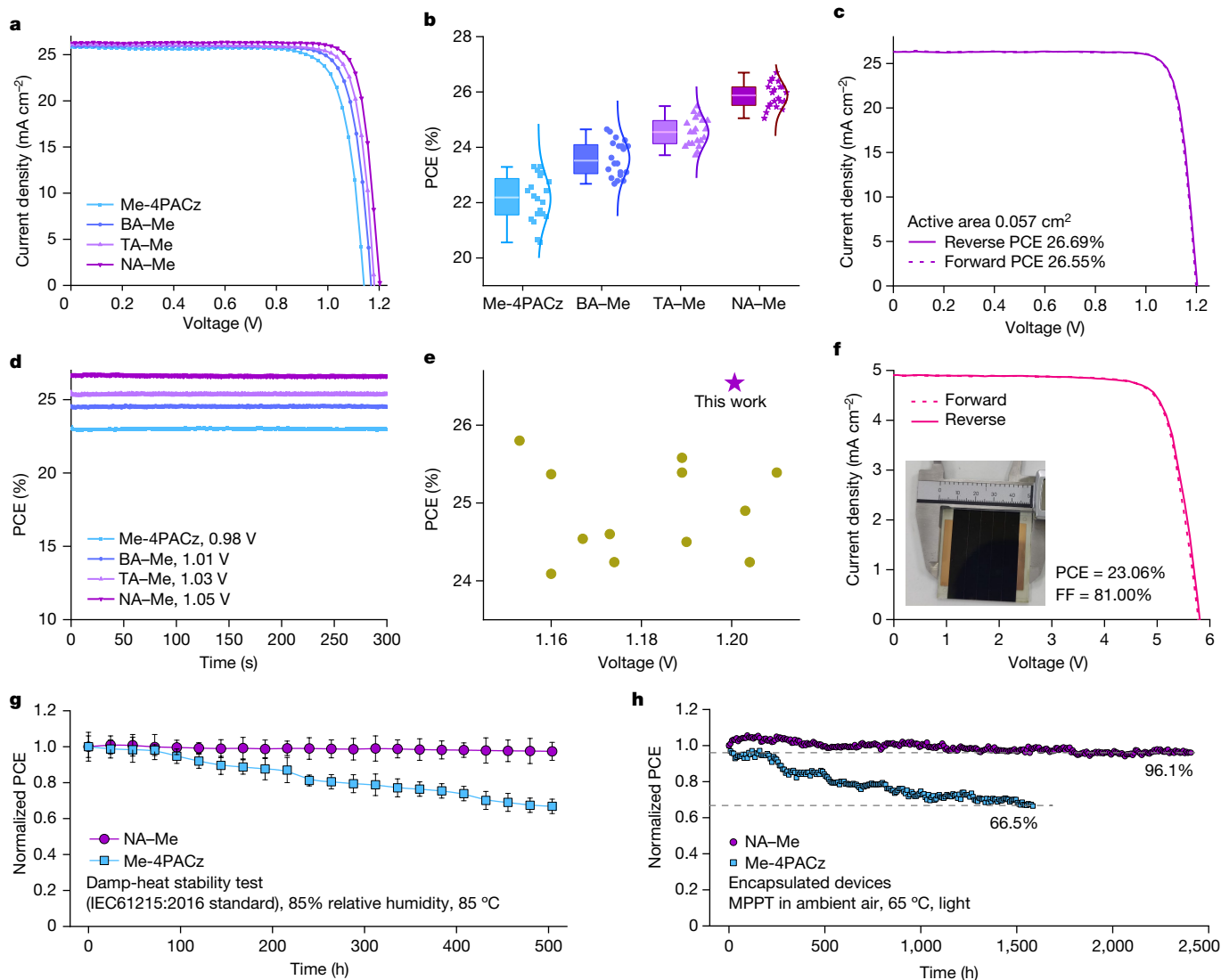


Fig. 4 | Photovoltaic performance of devices. **a**, J - V curves of PSCs with different SAMs under reverse scan. **b**, Statistics of the PCE of PSCs with different SAMs based on 20 devices. **c**, J - V curves of the champion inverted PSC with the NA-Me HSL during reverse and forward scans. **d**, Stabilized PCEs of champion inverted PSCs based on different SAMs determined by MPPT for 300 s. **e**, Comparison of certified PCE and V_{OC} of our PSCs with other reported high-performance inverted PSCs. **f**, J - V curves of champion mini-modules based on the NA-Me HSL. The inset shows a photograph of the mini-module

which accounts for about 93% of the theoretical thermodynamic maximum for a material with this bandgap^{28,29}. The integrated J_{SC} value for the external quantum efficiency (EQE) of the champion device with NA-Me SAM is 25.57 mA cm⁻² (Supplementary Fig. 37). One such device was measured by an independently accredited testing centre (Chinese National PV Industry Measurement and Testing Center), yielding a PCE of 26.54% under reverse scan (Supplementary Fig. 38) and a record steady-state PCE of 26.54% after maximum power point tracking (MPPT) for 300 s (Supplementary Fig. 39), which is one of the highest certified efficiencies among all single-junction PSCs (Fig. 4e and Supplementary Table 8). This result confirms the superiority of our hybrid SAM strategy (Supplementary Table 9 and Supplementary Note 11).

The FF loss analysis (Supplementary Note 12 and Supplementary Figs. 40 and 41) showed minimized non-radiative recombination loss in the NA-Me sample, suggesting that the defect-assisted trap states at the buried interface were suppressed, consistent with our DFT calculations. Furthermore, based on the results of transient photovoltage and

with an aperture area of 11.1 cm². **g**, Normalized PCE of encapsulated devices aged under damp heat conditions (85 °C and 85% relative humidity) following the IEC61215:2016 standard. The initial PCEs of the Me-4PACz and NA-Me devices used for the damp heat tests were 22.68% and 26.15%, respectively. **h**, Operational stability of encapsulated devices under continuous 1-sun-equivalent white-light LED illumination with MPPT in ambient air (65 °C). The initial PCEs of the Me-4PACz and NA-Me devices used for the operational stability tests were 22.75% and 26.21%, respectively.

capacitance-voltage (C - V) measurements (Supplementary Figs. 42 and 43), we conclude that NA-Me SAMs can effectively suppress non-radiative recombination and improve charge transfer at the buried interface³⁰. We also considered the upscaling capability of the NA-Me hybrid HSL and successfully fabricated high-performance mini-modules (Fig. 4f). The mini-module with an aperture area of 11.1 cm² exhibited a PCE of 23.06% (Supplementary Table 10) from the reverse J - V scan with low hysteresis. We sent one of the mini-modules to an independent laboratory (Shanghai Institute of Microsystem and Information Technology) for certification, and a PCE of 22.74% was obtained from the reverse J - V scan (PCE of 22.58% from the forward scan; Supplementary Fig. 44). To the best of our knowledge, this result represents the highest reported certified efficiency of inverted perovskite mini-modules to date (Supplementary Fig. 45 and Supplementary Table 11).

We further estimated the effect of the NA-Me SAM on the perovskite and device stability. The stabilities of the perovskite films with different SAMs (Me-4PACz and NA-Me) were evaluated using in situ

photoluminescence measurements in ambient air (Supplementary Fig. 46). The photoluminescence intensity of the perovskite film on Me-4PACz decreased after continuous laser irradiation for 120 min at 85 °C, whereas the photoluminescence intensity of the perovskite film on NA-Me showed no significant change. This observation indicates that NA-Me can effectively enhance the thermal and light stabilities of perovskite films. We suggest that amorphous buried interfaces with numerous defects and high residual stresses are often more prone to decomposition during ageing^{14,31} (Supplementary Fig. 47). As the incident side of light, the buried interfaces with improved quality would improve device stability.

The efficiency evolutions of the inverted PSCs under stresses were further monitored. It should be noted that the Bi/Ag bilayer was used to replace Ag as the electrode to fabricate devices with the configuration of ITO/NiO/Me-4PACz or NA-Me/perovskite/PI/PCBM/BCP/Bi/Ag for stability study because of the inferior operational stability of Ag-based devices³² (Supplementary Fig. 48). First, we placed encapsulated devices in a chamber (Supplementary Fig. 49) to conduct the damp heat test (85 °C and 85% relative humidity) following the International Electrotechnical Commission (IEC) 61215:2016 standard. As shown in Fig. 4g, the NA-Me-based devices maintained 97.4% of their initial PCEs (26.15%; Supplementary Fig. 50a) over 500 h under damp heat ageing, whereas the devices with Me-4PACz alone retained only 66.8% of their initial PCEs (22.68%; Supplementary Fig. 50b) within the same ageing time. Moreover, we also examined the operational stability of the encapsulated devices with MPPT under continuous 1-sun-equivalent white-light LED illumination at 65 °C hot plate in ambient air (Fig. 4h). The NA-Me-based device retained approximately 96.1% of its initial PCE (26.21%; Supplementary Fig. 51a) over 2,400 h, whereas the device with Me-4PACz retained only 66.5% of its initial PCE (22.75%; Supplementary Fig. 51b) after 1,580 h. Therefore, our devices achieved both high efficiency and good operational stability (Supplementary Table 12), which demonstrates that our interface molecular hybrid engineering using NA-Me is an effective approach for substantially improving the efficiency and stability of inverted PSCs.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-024-07723-3>.

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Methods

Materials

All the solvents and reagents were purchased and used directly without further purification. Nickel oxide (NiO) nanoparticles, organic halide salts (formamidinium iodide (FAI), lead iodide (PbI₂) and [6,6]-phenyl-C₆₁-butyric acid methyl ester (PC₆₁BM) were purchased from Advanced Election Technology. Methylamine hydrochloride (MACl), [4-(3,6-dimethyl-9*H*-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz), caesium iodide (CsI) and bathocuproine (BCP) were obtained from TCI America. Piperazinium iodide (PI) was purchased from Xi'an Yuri Solar. Benzoic acid (BA), trimesic acid (TA) and 4,4',4''-nitrilotribenzoic acid (NA) were purchased from Aladdin Biochemical Technology. Unless stated otherwise, all solvents and reagents were purchased from Sigma-Aldrich and used as received.

Perovskite precursor and film preparation

For the composition FA_{0.95}Cs_{0.05}PbI₃, 1.5 M perovskite precursor solution was prepared by mixing CsI, FAI and PbI₂ in DMF:DMSO = 4:1 (vol/vol) mixed solvent subject to the stoichiometric ratio. Additional 3 mol% PbI₂ and 10 mol% MACl were added to the precursor for better crystallization and perovskite phase transformation. The perovskite precursor (60 μ l) was spin-coated at 4,000 rpm for 50 s (5 s acceleration to 4,000 rpm). Ethyl acetate (200 μ l) was dropped on the film at 20 s before the end of the spinning. The film was immediately annealed at 100 °C for 30 min.

Small-area devices fabrication

Glass/ITO substrates were cleaned thoroughly by sequential ultrasonication for 20 min in a detergent solution, distilled water, alcohol and acetone. Then, the substrates were dried with N₂ and cleaned with UV ozone for 30 min before use. The NiO hole transport layer was formed by spin-coating NiO nanoparticle ink with a concentration of 10 mg ml⁻¹ in a mixed solution of deionized water and IPA (3:1, vol/vol) onto the ITO substrate at 3,000 rpm for 20 s, then annealed at 110 °C for 10 min in an air environment and then transferred to an N₂-filled glovebox. Subsequently, the solutions of Me-4PACz and candidate molecules (BA, TA and NA) with the same concentration (1 mmol l⁻¹ in absolute ethanol) were mixed in a molar ratio of 3:1 in characterization and device fabrication and then were spin-coated on the NiO film at 3,000 rpm for 20 s, followed by annealing at 100 °C for 10 min to form mixed Me-4PACz HSLs. The perovskite films were spin-coated on NiO/mixed HSLs (or NiO/Me-4PACz) following the abovementioned method. For the interfacial passivation layer, PI with the concentration of 0.4 mg ml⁻¹ in IPA was spin-coated on top of perovskite films (5,000 rpm, 30 s) and annealed at 100 °C for 5 min. Then, the PCBM solution (20 mg ml⁻¹, in chlorobenzene) was spin-coated on top of the interfacial passivation layer as an electron transport layer at 3,000 rpm for 30 s and annealed at 70 °C for 10 min. Sequentially, the saturated solution of BCP in methanol was dynamically spin-coated on the PCBM layer at 6,000 rpm for 30 s and annealed at 70 °C for 10 min. The device was completed by thermally evaporating the electrode of 100 nm Ag with a controlled evaporation rate of 0.1 Å s⁻¹ to 0.5 Å s⁻¹, respectively, in a vacuum chamber (<3 × 10⁻⁴ Pa). The Bi/Ag (20 nm/100 nm) electrodes in the devices used for stability tests were thermally evaporated with a controlled evaporation rate of 0.1 Å s⁻¹ and 0.1–0.5 Å s⁻¹, respectively.

Mini-modules fabrication

The mini-modules were fabricated on a glass/FTO substrate of size 5 cm × 5 cm. The cleaned glass/FTO substrate was subjected to laser-scribing using a 532-nm nanosecond laser machine with a power of 2 W to form P1 lines-isolated cell units. The substrates were dried with N₂ and cleaned in UV ozone for 30 min before use. FA_{0.95}Cs_{0.05}PbI₃-based perovskite precursor (300 μ l) was spin-coated onto the substrates

(5 cm × 5 cm) at 4,000 rpm for 50 s (5 s acceleration to 4,000 rpm). Ethyl acetate (500 μ l) was slowly dropped on the film at 20 s before the end of the spinning. After depositing the BCP layer, the module was re-etched to form P2 lines with a power of 0.3 W. Finally, it formed effective monolithically interconnected modules by scribing the Ag (100 nm) electrode using a power of 0.3 W to form P3 lines. P4 is an etching procedure for cleaning the edge of the modules; the laser used in P4 is the same as in P1. The fabricated modules typically have five sub-cells, and each sub-cell has a width of 6.8 mm. The total scribing line width was about 0.17 mm, giving a GFF of around 97.5% with an aperture area of 11.1 cm².

Characterization

The XRD characterizations were recorded on an Empyrean X-ray diffractometer with Cu K α radiation (PANalytical). The top-view and the cross-sectional images of the samples were obtained by a field-emission SEM (FEI NOVA NanoSEM 450). Ultraviolet–visible absorption spectra were characterized on an ultraviolet–visible spectrometer (PerkinElmer). The GIWAXS measurement was carried out with the Xeuss 3.0. The incident angle was 0.5°. The GIXRD patterns were acquired in the air by using a Rigaku Smartlab with Cu K α radiation in the 2 θ range of 30.6°–32.6°. The steady-state photoluminescence spectra were recorded by the Edinburgh FLS920 fluorescence spectrometer (Edinburgh) with an excitation source wavelength of 532 nm. The TRPL spectrum was obtained by a fluorescence spectrometer with an excitation wavelength of 532 nm (DeltaFlex, HORIBA). Fluorescence lifetime imaging microscopy was conducted using FLIM300 (Dalian Chuangrui Spectral Technology Co., Ltd). The UPS spectra were investigated by an AXIS-ULTRA DLD-600 W Ultra spectrometer (Kratos) with a non-monochromated He I α photon source ($h\nu$ = 21.22 eV). The water contact angle was recorded by Drop Shape Analyser DSA100 from KRÜSS. The in situ photoluminescence spectra measurement was performed by a laser with the wavelength of 532 nm, and this device was fixed on a micro-spectral measurement system, ProSp-Micro40-VIS (Hangzhou SPL Photonics), together with a spectrograph (Ocean Insight). An XPS system (Thermo ESCALAB 250XI) equipped with a monochromatic Al K α X-ray source (1486.6 eV) operating at 100 W was used to acquire the XPS spectra. KPFM measurements were obtained using a dimension icon scanning probe microscope (Asylum Research, MFP-3D-Stand Alone) in the tapping mode. The J – V curves of the devices were recorded with a black mask with an active area of 0.057 cm² by a Keithley 2400 source measurement unit and a solar simulator (Oriel, model 9119) with an AM1.5 G spectrum. The light intensity was calibrated with a reference silicon solar cell (KG3, Enlitech). The scanning speed for both forward and reverse scans was 0.01 V s⁻¹. The EQE spectra were measured under d.c. mode with the EQE system (Saifan) using monochromatic light of 1 × 10¹⁶ photons per cm⁻². The transient photovoltage measurements and Mott–Schottky plots were obtained by a Zennium electrochemistry workstation (Zahner). The PLQY of the corresponding film was measured on QuantaMaster 8000 (HORIBA) with a 532-nm laser to photoexcite the samples placed in an integrating sphere. The intensity of the laser was calibrated by adjusting the power when it produced a 1-sun-equivalent current density from the devices at J_{sc} condition.

Stability tests

The encapsulation of the PSC was processed according to our previous report³³. After the corresponding encapsulation process, operational stability tests were performed for the fabricated devices. The operational stability of the encapsulated devices was measured by a commercial multichannel stability test system (Wuhan 91PVKSolar) operating in the MPP tracking mode at a temperature of 65 °C. The illumination was achieved by a 1-sun-equivalent white-light LED and the light intensity was calibrated to achieve the same J_{sc} from the devices measured under a standard solar simulator (AM 1.5 G, 100 mW cm⁻²).

For the damp heat test, we put the encapsulated devices in the environmental test chamber at 85 °C/85% relative humidity and measured the PCEs periodically after the devices cooled to room temperature.

Molecular dynamics simulations

Classical molecular dynamics simulations for the perovskite/SAMs/NiO heterojunction used the LAMMPS program³⁴. The equation of motions was integrated with a time step of 1 fs using the velocity Verlet method³⁵. The trajectories were propagated in a canonical ensemble (NVT) with the Nosé–Hoover thermostat^{36,37}, with a time constant of 0.1 fs. The perovskite and NiO surfaces were generated by the $16 \times 17 \times 3$ supercell of FAPbI₃ and the $25 \times 26 \times 2$ supercell of NiO and the SAMs layer is set to 20 Å. The SAMs layer in four heterojunction models contains the following: (1) 90 Me-4PACz molecules; (2) 90 Me-4PACz and 30 BA molecules; (3) 90 Me-4PACz and 30 NA molecules; and (4) 90 Me-4PACz and 30 TA molecules, which were randomly generated between the surfaces by the Packmol program³⁸. Periodic boundary conditions were applied to the *x*- and *y*-directions, whereas two top perovskite layers and the bottom NiO layers were fixed. The velocities were initialized at 400 K and equilibrated in 10 ps, cooled to 298.15 K in the subsequent 10 ps, and then further equilibrated at 298.15 K in another 10 ps. The production run took 0.5 ns at 298.15 K. The surface adsorption analysis used our in-house code, which assigned SAMs to adjacent surfaces and computed the surface coverage by searching any atoms of SAMs above the surface within a distance threshold of 5 Å and a radial threshold of 2 Å. The interactions of the perovskite surface used the recently developed transferable perovskite force field³⁹. The interactions of SAMs used the DREIDING force fields⁴⁰, in which the restrained electrostatic potential charges⁴¹ were computed at PBE0/cc-pVDZ level using the ORCA 5.0.3 program⁴². The interactions between the perovskite and SAMs used the universal force field⁴³. The NiO surface was described by the INTERFACE force field⁴⁴, which showed good compatibility with the DREIDING force field for simulating molecular adsorptions on metal oxides. The interactions between the perovskite and NiO surface were ignored. The molecular dynamics simulations for the heterojunctions with perovskite surface defects used the same setting as for the adsorption simulation, except that we created a charged Pb vacancy on the perovskite surface.

DFT calculations

The geometry optimizations for surface adsorptions, molecular clusters, intrinsic surface defects and passivated surface defects used the VASP code⁴⁵ with a plane wave (PW) basis set and projected augmented wave (PAW) pseudopotentials (PPs). We used the PBE flavour of the GGA PPs, and long-range dispersion interactions were considered by the DFTD3 scheme. The kinetic energy cutoff is 500 eV for the expansion of the wavefunctions. The convergence thresholds were set to 1.0×10^{-6} eV for energy and $-0.05 \text{ eV } \text{\AA}^{-1}$ for the norm of the atomic forces. The adsorption calculations on the (100) surface of the perovskite and NiO used the $3 \times 3 \times 2$ FAPbI₃ and $5 \times 5 \times 2$ NiO slabs, respectively. The Hubbard corrections for Ni used $U = 8 \text{ eV}$ and $J = 0.95 \text{ eV}$. The perovskite surface defect calculations used the $3 \times 3 \times 2$ FAPbI₃ slab, including the V_{I} , V_{Pb} , Pb_i and I_pb defects with various point charges. A 17 Å and 14 Å vacuum space were added to the perovskite and NiO slabs, respectively. Dipole corrections were applied in the *z*-direction for all adsorption calculations. All ionic optimizations were done at the gamma point, and the bottom two atom layers were fixed during the optimizations. The charge densities of the perovskite defects were recomputed with a $2 \times 2 \times 1$ k-point mesh centred at the gamma point for band structure calculations, including 40 points in the two-dimensional k-space.

Data availability

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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Author contributions W.C., N.-G.P., Z.L. and S. Liu conceived the project; S. Liu, Z.S., R.C. and Y. Zhang fabricated the devices; S. Liu, W.X., Z.S., Y. Zhang, R.C., J.W., F.R., Q.Z. and H.R. carried out the material and device characterizations; S.H. and M.K.-C. conducted the modelling of PLQE dependent on light intensity (two-trap-level SRH model) under the supervision of H.J.S.; S. Li, H.L. and L.Q. carried out the GIXRD measurements; Y.J. and W.H. carried out the GIWAXS measurements; R.C., Y. Zhang, J.W., F.R., Q.Z., H.R., Y.J., W.H. and Y.G. provided suggestions for writing the paper; W.X. performed the DFT calculations under the supervision of Y. Zhao; X.L. performed the DFT calculations under the supervision of J.L. and Y. Zhang; J.L. performed the molecular dynamics simulations and analysed all computational data; S. Liu, J.L., Z.S., R.C., B.X., Z.L. and W.C. analysed all experimental data; and S. Liu, J.L., W.X., R.C., Z.S. and Y. Zhang prepared the paper under the supervision of Z.L., N.-G.P. and W.C. All authors revised and proofread the written paper.

Competing interests H.J.S. is a co-founder and the Chief Scientific Officer of Oxford PV, a company commercializing perovskite solar cells.

Additional information

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Correspondence and requests for materials should be addressed to Zonghao Liu, Nam-Gyu Park or Wei Chen.

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